

Superposition Coding Beyond Binary Alphabets at Finite Blocklength: 4-Point and 8-Point Constellations

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Technical report accompanying the BSBC polar superposition paper

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Abstract

The companion conference paper finds that on the binary symmetric broadcast channel (BSBC), superposition coding (SUP) does not outperform time-division multiplexing (TDM) until about 2^{11} channel uses, even with random coding, and conjectures that larger alphabets could be more favorable. This report tests that conjecture with normal approximations and, where noted, polar-code simulations. For 2-bit (4-point) symbols it does not hold. With power-domain superposition on the AWGN channel, the best 4-point constellation is the orthogonal I/Q split, which equals TDM with power control. With discrete-domain superposition, the quaternary symmetric broadcast channel needs 2^{10} – 2^{12} symbols before SUP beats TDM, and hard-decision Gray QPSK is exactly two BSBC uses per symbol, reproducing the paper. For 3-bit (8-point) symbols the picture changes. Hard-decision Gray 8PSK and rectangular 8QAM protect their label bits unequally, and splitting the bits between the users (the weak user takes the robust bits) beats TDM, even TDM with power control, by 0.03–0.22 bits/symbol already at 128 symbols (0.07–0.25 at 8192). In this regime the paper’s construction (a biased satellite on every bit) is the wrong design, and each user’s layer is an ordinary point-to-point polar code. Mixing split modes across symbols, with one code per user, gives continuous control of the rate split and is within 0.012 bits/symbol of the best design found at $n = 128$. Polar-code simulations at $n = 128$ (SC decoding, block error rate $\leq 10^{-2}$) confirm this: mode mixing lies above the convex hull of all simulated TDM points by 0.01–0.23 bits/symbol, while the paper’s construction, even with a well-chosen satellite selection rule, stays 0.06–0.61 bits/symbol below it. The same simulator reproduces or improves on the thesis’s realized polar superposition points on the BSBC.

1 Summary of findings

1. **Power domain, 4-point constellation.** Over the constellations $x(v, s) = \sqrt{1-\beta}(-1)^v + \sqrt{\beta}e^{j\theta}(-1)^s$, the optimum is $\theta = 90^\circ$ at every SNR pair and blocklength evaluated. This I/Q split is identical to TDM with $\lambda = \frac{1}{2}$ and power control (Section 4.1), so the gain over TDM with power control is zero to within ± 0.008 bits/symbol. Polar codes at $n = 128$ agree.
2. **Discrete domain, quaternary symmetric BC (QSBC).** SUP first beats TDM at 2^{10} – 2^{12} symbols, with an asymptotic advantage of 0.06–0.08 bits/symbol (plus up to 0.026 from unstructured designs).
3. **Hard-decision Gray QPSK** is exactly two BSBC uses per symbol (Section 6). It reproduces the paper (crossover at 2^{10} symbols, asymptotic advantage 0.095 bits/symbol), and no

genuinely 4-ary design improves on it. The QSBC is a pessimistic model of hard-decision QPSK.

4. **Hard-decision 8PSK and 8QAM** (Section 8). Gray labels protect the three bits unequally. A pure bit split beats TDM with power control by 0.03–0.22 bits/symbol at $n = 128$ and 0.07–0.25 at $n = 8192$, depending on the channel. The paper’s construction loses 0.15–0.29 bits/symbol to equal-power TDM at $n = 128$.
5. **Bit-level polarization** (Section 9). Polar codes polarize each bit level at that level’s own rate; they do not move protection between levels. With unequal levels the modulation supplies the cloud/satellite split, each user’s code is an ordinary point-to-point polar code, and the satellite-selection problem of the paper does not arise.
6. **Fine rate control** (Section 10). Mixing two split modes across symbols, with one polar code per user spanning its bits, is within 0.006 (8PSK) and 0.012 (8QAM) bits/symbol of the best design at $n = 128$. A biased satellite on one level (the paper’s construction used as a knob) loses up to 0.22 bits/symbol at $n = 128$ and becomes competitive only from $n \approx 2048$.
7. **Polar simulations** (Section 11). At $n = 128$ and 13/8 dB with SC decoding, polar mode mixing beats the convex hull of all simulated polar TDM points (equal power and three power splits) at every rate split tested, by 0.01–0.23 (8PSK) and 0.05–0.21 (8QAM) bits/symbol. The polar codes lose 0.23–0.48 bits/symbol to the normal approximation. The paper’s construction on every bit stays below TDM (0.06–0.61 bits/symbol with the best selection rule); its satellite rate depends strongly on how the satellite positions are chosen. Exhaustive search at $n = 16$ and 32 and a local search at $n = 128$ show the best reliability rule is within one position, or 0.03–0.06 bits per channel use, of the best sets found.
8. **Against CCP**, SUP wins from about 32 QPSK symbols in both the binary and the quaternary model at equal SNR.

2 Background and question

The paper studies the two-user BSBC(p_1, p_2) with $0 < p_1 < p_2 < \frac{1}{2}$ and per-user reliability $\varepsilon_1 = \varepsilon_2 = 10^{-2}$. SUP uses a uniform cloud bit V and a satellite $X = V \oplus S$, $S \sim \text{Bern}(\alpha)$; receiver 2 decodes V treating S as noise, and receiver 1 uses successive interference cancellation (SIC). Its normal approximation (Proposition 1 of the paper, following [1, 2]) shows that for BSBC(0.01, 0.10) the SUP region extends beyond the TDM region only from $n = 2^{11}$, with an asymptotic advantage of 0.048 bits per use. Polar SUP codes [3] at $n = 128$ fall short of TDM accordingly, consistent with Gaussian results in which orthogonal access wins at short blocklengths and low spectral efficiency [4].

“Moving to a larger alphabet” admits several models, and they behave differently:

- *Power-domain superposition* (Section 4): soft-decision AWGN, uniform bits for both users, the satellite’s strength set by the constellation geometry.
- *Discrete-domain superposition* (Sections 5–6): a quaternary discrete memoryless channel, either the fully symmetric QSC or the hard-decision Gray QPSK channel, with the satellite’s strength set by how often it changes the cloud symbol. This is the direct 4-ary analogue of the paper’s construction.
- *Hard-decision 8-ary channels* (Sections 8–10): Gray 8PSK and rectangular 8QAM with nearest-point decisions, whose label bits are unequally protected.

3 Methods

Normal approximation. For each SUP design we compute the means and (co)variances of the information densities $\imath_2(V; Y_2)$, $\imath_c(V; Y_1)$ and $\imath_s(X; Y_1|V)$ and apply the per-user normal approximation of the paper’s Proposition 1: $R_2 \leq I_2 - \sqrt{V_2/n} Q^{-1}(\varepsilon_2)$, and at receiver 1 the bivariate Gaussian probability $F(\varepsilon_c, \varepsilon_s; \rho) \geq 1 - \varepsilon_1$ for the two SIC stages. Third-order terms are omitted for all schemes. Values for $n \leq 64$ are indicative only.

Baselines. TDM gives user 1 a fraction λ of the symbols, each user coding at its point-to-point normal-approximation rate with uniform input. With power control, user j ’s slots use energy P_j with $\lambda P_1 + (1 - \lambda)P_2 = 1$; for the hard-decision channels this means the same constellation and detector at SNR $P_j \text{SNR}_j$. CCP sends both messages in one codeword decoded by both receivers, $R_1 + R_2 \leq \min_j [C_j - \sqrt{V_j/n} Q^{-1}(\varepsilon_j)]$.

Metric. The advantage of SUP at blocklength n is $\max_{R_2} [R_1^{\text{SUP}}(R_2) - R_1^{\text{TDM}}(R_2)]$ over interior rate pairs, those where both schemes have $R_1 > 0.02$. It is negative when SUP is dominated. Its sign, and so the crossover blocklength, is robust; the size of a negative value depends on the interior definition, because both regions meet near their endpoints. (An earlier version required only $R_1^{\text{TDM}} > 0.02$, which lets SUP’s clipped rate of zero floor the value at about -0.02 .)

Numerics. AWGN moments use 2-D Gauss–Hermite quadrature (48^2 nodes), checked against 4×10^5 -sample Monte Carlo. Discrete-channel moments are exact by enumeration, checked against closed forms and 10^6 -sample Monte Carlo. The hard-decision 8PSK transition probabilities are exact radial integrals over each decision sector with Gauss–Legendre quadrature in angle; the rectangular 8QAM ones are products of 1-D Gaussian integrals; both agree with Monte Carlo to within 0.003. Polar codes use $G_n = F^{\otimes m}$, exact successive-cancellation (SC) decoding, and a genie-aided Monte-Carlo construction on each layer’s own channel; information sizes are the largest meeting a simulated block error rate of at most 10^{-2} .

Design checks. Each discrete study searches structured design families and then checks them with random and local searches over general cloud and satellite distributions. A design with cloud rate R_2^d also achieves every smaller R_2 , so the family boundary at R_2 is $\max\{R_1^d : R_2^d \geq R_2\}$, and a searched design’s excess is measured against that.

4 Power-domain superposition with a 4-point constellation

Model. Complex AWGN broadcast channel $Y_j = X + Z_j$, $Z_j \sim \mathcal{CN}(0, 1/\text{SNR}_j)$, $\text{SNR}_1 > \text{SNR}_2$. The cloud bit v (user 2) and the satellite bit s (user 1) select

$$x(v, s) = \sqrt{1 - \beta} (-1)^v + \sqrt{\beta} e^{j\theta} (-1)^s, \quad \mathbb{E}|X|^2 = 1.$$

$\theta = 90^\circ$ gives a rectangular 4QAM with the users on separate I/Q dimensions; $\theta = 0$ gives a one-dimensional, 4-PAM-like hierarchical constellation. TDM uses Gray QPSK in each user’s slots. One SNR pair, 7.33/2.15 dB, is chosen so that hard-decided Gray QPSK bits are exactly BSC(0.01) and BSC(0.10), i.e., the paper’s BSBC.

Table 1: Largest R_1 advantage (bits/symbol) of the best 4-point SUP constellation over TDM. “Finite n ” is the range over $n = 2^7, \dots, 2^{12}$. The optimum is $\theta = 90^\circ$ in every case.

SNR ₁ /SNR ₂ (dB)	QPSK capacities (user 1, user 2)	vs. equal-power TDM		vs. TDM with power control	
		$n \rightarrow \infty$	finite n	$n \rightarrow \infty$	finite n
7.33 / 2.15 (BSBC-matched)	1.921, 1.308	+0.078	+0.065 to +0.070	+0.005	−0.005 to −0.003
10 / 0	1.994, 0.972	+0.244	+0.241 to +0.248	0.000	+0.001 to +0.004
15 / 0	2.000, 0.972	+0.396	+0.396 to +0.449	0.000	0.000 to +0.008
20 / 5	2.000, 1.718	+0.130	+0.140 to +0.202	0.000	−0.003 to +0.004

4.1 The I/Q split is TDM with power control

At $\theta = 90^\circ$ the cloud occupies the in-phase dimension with energy $1 - \beta$ and the satellite the quadrature dimension with energy β . The noise has variance $1/(2 \text{SNR}_j)$ per dimension, so user 2 sees n uses of a binary-input AWGN channel at per-dimension SNR $2(1 - \beta)\text{SNR}_2$ and user 1, with no interference, n uses at $2\beta \text{SNR}_1$. TDM with $\lambda = \frac{1}{2}$ and powers $P_2 = 2(1 - \beta)$, $P_1 = 2\beta$ gives each user $n/2$ QPSK symbols, i.e., n binary-input dimensions at per-dimension SNR $P_j \text{SNR}_j$. The two schemes give each user the same channel with the same number of uses, so their rates, dispersions and polar codes coincide.

4.2 Normal-approximation results

Table 1 sweeps $\beta \in (0, 1)$ (64 values) and $\theta \in \{0, 15, \dots, 90\}^\circ$. At every SNR pair and blocklength, the best configuration is $\theta = 90^\circ$. Its advantage over equal-power TDM is present from $n = 128$ on, but against TDM with power control it vanishes.

Why $\theta = 90^\circ$ wins. With one bit per user per symbol, each user needs only one real dimension. The I/Q split gives each user a full dimension free of interference; any $\theta < 90^\circ$ makes the users share a dimension and interfere.

4.3 Polar codes at $n = 128$

Figure 1 shows realized rate pairs at the BSBC-matched SNRs with SC decoding and per-user block error rate $\leq 10^{-2}$. Every $\theta < 90^\circ$ point lies below the $\theta = 90^\circ$ point with the same β . The $\theta = 90^\circ$, $\beta = 0.5$ code realizes (0.398, 0.852), matching equal-power TDM at $\lambda = \frac{1}{2}$, (0.391, 0.859), as Section 4.1 predicts; the other $\theta = 90^\circ$ points track the power-controlled TDM points within the simulation’s resolution. The two layers carry independent uniform bits, so each is an ordinary point-to-point polar code; this is the polar-coded NOMA setting of [5].

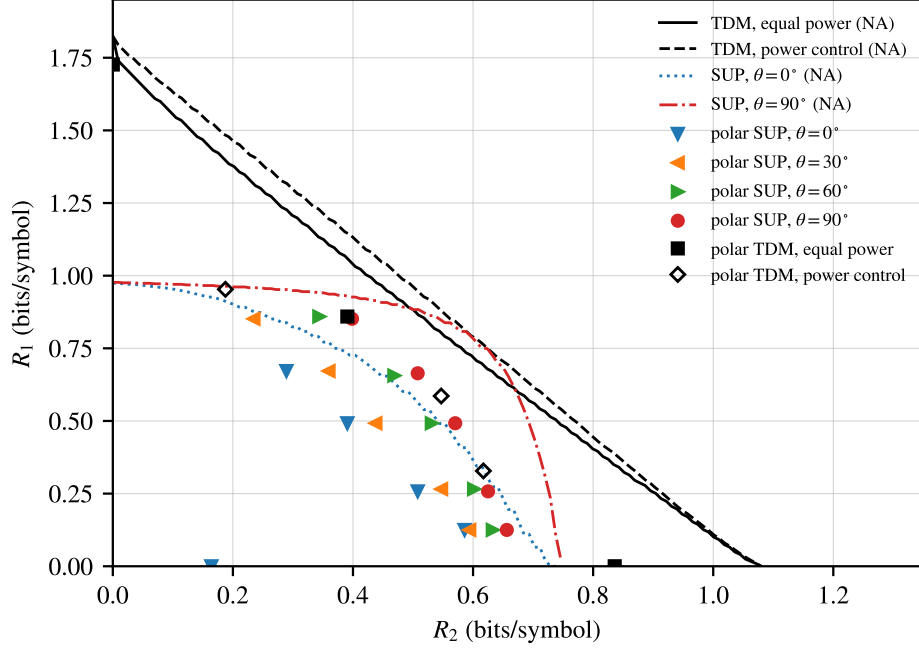


Figure 1: 4-point AWGN broadcast channel at 7.33/2.15 dB, $n = 128$ symbols, $\varepsilon_1 = \varepsilon_2 = 10^{-2}$: normal approximations (lines) and polar codes with SC decoding (markers).

5 Discrete domain: the quaternary symmetric BC

Model. QSCB(p_1, p_2): $Y_j = X + Z_j \pmod{4}$ with $P(Z_j = 0) = 1 - p_j$ and $P(Z_j = z) = p_j/3$ for $z \neq 0$. The point-to-point capacity is $C_{\text{QSC}}(p) = 2 - h(p) - p \log_2 3$ and the dispersion $V_{\text{QSC}}(p) = p(1 - p) \log_2^2 \frac{3(1-p)}{p}$. Two QSCs in series form a QSC, so the channel is degraded. Matching the symbol error rate of hard-decided QPSK at the BSBC-matched SNRs gives $p_j = 1 - (1 - q_j)^2$ with $q_j \in \{0.01, 0.10\}$, i.e., QSCB(0.0199, 0.19).

Superposition families. The cloud V is uniform on $\mathbb{Z}_4$ and $X = V + S \pmod{4}$, with *symmetric* S : $P(S = 0) = 1 - a$, $P(S = s) = a/3$; *subgroup* $S \in \{0, 2\}$ with $P(S = 2) = a$; and *bit-layer*: a binary cloud selects the pair $\{V, V + 2\}$ and the satellite picks $V + 2$ with probability a . A random search over general $(P_V, P_{X|V})$ with $|\mathcal{V}| = 4$ checks whether the families reach the asymptotic boundary.

Results. Table 2 gives the asymptotic and finite- n advantage. The symmetric and subgroup families form the boundary; general distributions exceed them by up to 0.026 bits/symbol, so the true advantage is somewhat larger and the crossovers may come somewhat earlier than shown.

6 Hard-decision Gray QPSK as a discrete channel

Model. Label the QPSK points by their Gray bits (b_I, b_Q) : $00 \rightarrow (+, +)$, $01 \rightarrow (+, -)$, $11 \rightarrow (-, -)$, $10 \rightarrow (-, +)$, so neighbouring points differ in one bit. With hard decisions each bit is flipped independently with probability $q = Q(\sqrt{\text{SNR}})$, so

$$P(y | x) = (1 - q)^{2-d(x,y)} q^{d(x,y)},$$

Table 2: QSBC: largest R_1 advantage (bits/symbol) of SUP over TDM. “RS” is the further asymptotic gain found by random search. Finite- n values are normal approximations over interior rate pairs; only their sign and the crossover are robust.

	$n \rightarrow \infty$	RS	2^7	2^8	2^9	2^{10}	2^{11}	2^{12}	2^{13}
QSBC(0.0199, 0.19)	0.079	+0.020	-0.024	-0.036	-0.027	-0.004	+0.019	+0.036	+0.050
QSBC(0.01, 0.10)	0.069	+0.026	-0.018	-0.024	-0.033	-0.018	-0.001	+0.018	+0.033
QSBC(0.05, 0.30)	0.060	+0.019	< 0	< 0	< 0	+0.003	+0.020	+0.032	+0.042

Table 3: Transition probabilities from label 00, hard-decision Gray QPSK versus the QSC with the same symbol error rate.

to	Gray, $q = 0.01$	QSC(0.0199)	Gray, $q = 0.10$	QSC(0.19)
00 (correct)	0.9801	0.9801	0.81	0.81
01 or 10 (one bit wrong)	0.0099 each	0.0066 each	0.09 each	0.0633 each
11 (both bits wrong)	0.0001	0.0066	0.01	0.0633
capacity (bits/symbol)	1.838	1.828	1.062	0.997

with d the Hamming distance between the labels: $Y = X \oplus Z$ on $\mathbb{Z}_2^2$ with $Z = (\text{Bern}(q), \text{Bern}(q))$. This is two independent BSC uses per symbol, with capacity $2(1 - h(q))$ and dispersion $2V_{\text{BSC}}(q)$. Table 3 compares it with the QSC of the same symbol error rate: the QSC moves probability onto the diagonal symbol, which Gray QPSK almost never produces.

Superposition designs. Cloud uniform on the four labels and $X = V \oplus S$ with the bits of S independent, $P(S_I = 1) = a_I$, $P(S_Q = 1) = a_Q$ ($a_I = a_Q$ is the paper’s construction on both bits; $a_I = 0$, $a_Q = \frac{1}{2}$ is the I/Q split), plus symmetric and double-flip satellites and a random search over general distributions.

Results. The per-symbol moments with $a_I = a_Q = \alpha$ equal exactly twice the BSBC per-use moments, and the asymptotic advantage over TDM is 0.095 bits/symbol (2×0.048). The random search adds only 0.0025, i.e., no genuinely 4-ary design improves on the paper’s construction applied per bit. Table 4 gives the finite- n comparison: the Gray model crosses TDM at 2^{10} symbols, exactly as the binary analysis, and earlier than the QSC model. Below the crossover the best Gray design gives the weak user one bit alone and the strong user mostly the other, which amounts to TDM at $\lambda = \frac{1}{2}$; the paper’s single- α design cannot vary α between the bits and falls further behind.

7 Binary versus quaternary at equal SNR

Figure 2 compares the SUP advantage over TDM in QPSK symbols. In the binary model one QPSK symbol is two BSBC uses, so the BSBC curve is plotted at half its blocklength and twice its per-use rate. SUP first beats TDM at about 2^{10} symbols in the binary (and Gray-QPSK) model and at 2^{11} in the QSC model. The binary model’s asymptotic advantage (0.096 bits/symbol) also exceeds the QSC one (0.079, or up to about 0.10 with optimized satellite distributions). A quaternary alphabet therefore does not shorten the blocklength at which superposition pays off.

Table 4: Largest R_1 advantage of SUP over TDM (bits/symbol, $\varepsilon = 10^{-2}$) for hard-decision Gray QPSK, the paper’s BSBC at twice the blocklength, and the QSC model.

n (symbols)	2^7	2^8	2^9	2^{10}	2^{11}	2^{12}	2^{13}	2^{14}
Gray QPSK, best design	−0.007	−0.008	−0.004	+0.015	+0.037	+0.055	+0.064	+0.076
$2 \times$ BSBC(0.01, 0.10)	−0.109	−0.052	−0.017	+0.015	+0.040	+0.056	+0.067	+0.075
QSBC(0.0199, 0.19)	−0.024	−0.036	−0.027	−0.004	+0.019	+0.036	+0.050	+0.057

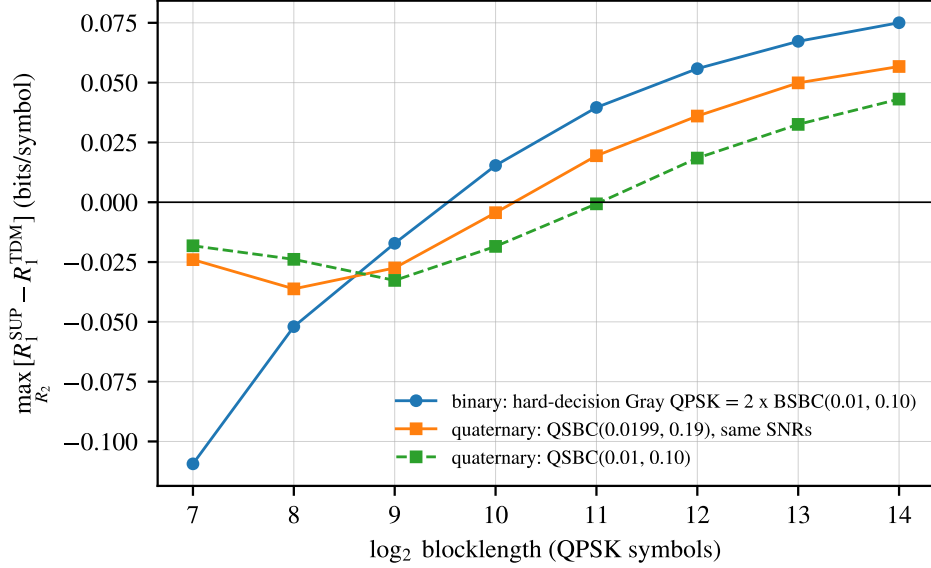


Figure 2: Largest R_1 advantage of SUP over TDM at equal R_2 (normal approximation, $\varepsilon_1 = \varepsilon_2 = 10^{-2}$, interior rate pairs), in bits per QPSK symbol, against blocklength in QPSK symbols ($\geq 2^7$).

Against CCP. CCP is a weaker baseline, and SUP overtakes it much earlier (Table 5). At equal SNR both models cross at 32 QPSK symbols (2^6 BSBC uses versus 2^5 QSBC symbols). These blocklengths are short enough that the normal approximation is only indicative; < 0 means no advantage.

Table 5: Largest R_1 advantage of SUP over CCP (normal approximation, $\varepsilon = 10^{-2}$), by blocklength in channel uses of each model.

	2^5	2^6	2^7	2^8	2^9	2^{10}
BSBC(0.01, 0.10) (binary uses)	< 0	+0.043	+0.214	+0.307	+0.352	+0.372
QSBC(0.0199, 0.19) (symbols)	+0.349	+0.387	+0.625	+0.733	+0.787	+0.809
QSBC(0.01, 0.10) (symbols)	+0.234	+0.231	+0.402	+0.487	+0.524	+0.534

8 Hard-decision 8PSK and 8QAM

Models. Complex AWGN with $E_s = 1$, nearest-point hard decisions and Gray labels (b_2, b_1, b_0) . *8PSK*: points $e^{j2\pi k/8}$ with label $k \oplus \lfloor k/2 \rfloor$; $P(y|x)$ is the Gaussian mass in the 45° sector of y . *8QAM*: the rectangular 4×2 grid $I \in \{\pm 1, \pm 3\}$, $Q \in \{\pm 1\}$ scaled to unit energy, a Gray 4-PAM label (b_2, b_1) on I and b_0 on Q ; $P(y|x)$ factors into I and Q terms. Unlike Gray QPSK, the three bits are unequally protected (Table 6), and they are not independent channels. Some SNR pairs are not stochastically degraded (8PSK at 13/8 dB and all 8QAM pairs, by margins of 10^{-8} – 4×10^{-3} in the degrading matrix); the SUP regions remain achievable because receiver 1 decodes the cloud explicitly.

Designs. Cloud uniform on the eight labels and $X = V \oplus S$, where the bits of S are independent with $P(S_b = 1) = a_b$ on an 8-point grid per bit (512 designs): $a_b = 0$ keeps bit b for the weak user, $a_b = \frac{1}{2}$ gives it to the strong user, and intermediate values put the paper’s construction on that bit. A symmetric 8-ary satellite, random designs and local searches complete the set.

Results. Table 7 and Figure 3. In every case the best design at finite blocklength is a pure bit split, i.e., discrete hierarchical modulation [6]: for 8PSK the weak user takes b_2b_1 and the strong user b_0 ; for 8QAM the weak user takes the sign bit b_2 and the strong user b_1b_0 . It beats TDM, and still beats TDM with power control, from about 64–128 symbols. The paper’s construction (the same a on every bit) is far behind at short blocklengths and overtakes equal-power TDM only at 2^{11} – 2^{13} symbols in four of the six cases, and not at all by 2^{13} at 7.33/2.15 dB. The local search finds designs up to 0.06–0.11 bits/symbol beyond the grid asymptotically, so the gains shown are conservative.

Why 8-ary differs from 4-ary. In Gray QPSK the two bits are equally protected and independent, so a bit split is exactly TDM at $\lambda = \frac{1}{2}$. In Gray 8PSK and 8QAM a bit split gives the weak user the bits that its own channel protects best and the strong user the bit that only it can resolve, which no orthogonal schedule reproduces, even with power control. In power-domain terms, the 8-ary constellation is itself a superposition of a coarse and a fine constellation.

Table 6: Bit error rates of the hard-decision Gray labels and point-to-point capacities (uniform input).

	SNR (dB)	bit error rate (b_2, b_1, b_0)	capacity (bits/symbol)
8PSK	13 / 8	(0.004, 0.004, 0.008) / (0.044, 0.044, 0.087)	2.868 / 2.150
8PSK	16 / 6	(0.0002, 0.0002, 0.0003) / (0.072, 0.072, 0.143)	2.992 / 1.807
8PSK	7.33 / 2.15	(0.053, 0.053, 0.105) / (0.146, 0.146, 0.268)	2.035 / 1.143
8QAM	13 / 8	(0.003, 0.005, 0.005) / (0.037, 0.074, 0.074)	2.887 / 2.057
8QAM	16 / 6	(0.0001, 0.0001, 0.0001) / (0.063, 0.125, 0.125)	2.995 / 1.654
8QAM	7.33 / 2.15	(0.045, 0.090, 0.090) / (0.122, 0.236, 0.230)	1.917 / 1.032

Table 7: Hard-decision 8PSK and 8QAM: largest R_1 advantage (bits/symbol, $\varepsilon = 10^{-2}$) of the best design over TDM with power control (“pc”) and equal-power TDM (“eq”), and of the paper’s construction (same a on every bit) over equal-power TDM.

	SNR (dB)	$n \rightarrow \infty$		best design vs. TDM with power control				paper’s construction vs. eq		
		eq	pc	2^7	2^9	2^{11}	2^{13}	2^7	2^{11}	2^{13}
8PSK	7.33 / 2.15	+0.156	+0.128	+0.032	+0.074	+0.097	+0.112	−0.151	−0.021	−0.005
8PSK	13 / 8	+0.155	+0.119	+0.130	+0.118	+0.127	+0.128	−0.232	−0.031	+0.008
8PSK	16 / 6	+0.285	+0.148	+0.195	+0.172	+0.170	+0.147	−0.287	−0.019	+0.039
8QAM	7.33 / 2.15	+0.259	+0.253	+0.222	+0.237	+0.245	+0.253	−0.153	−0.031	−0.024
8QAM	13 / 8	+0.171	+0.104	+0.145	+0.116	+0.081	+0.090	−0.204	+0.008	+0.055
8QAM	16 / 6	+0.319	+0.089	+0.145	+0.087	+0.060	+0.069	−0.257	+0.042	+0.086

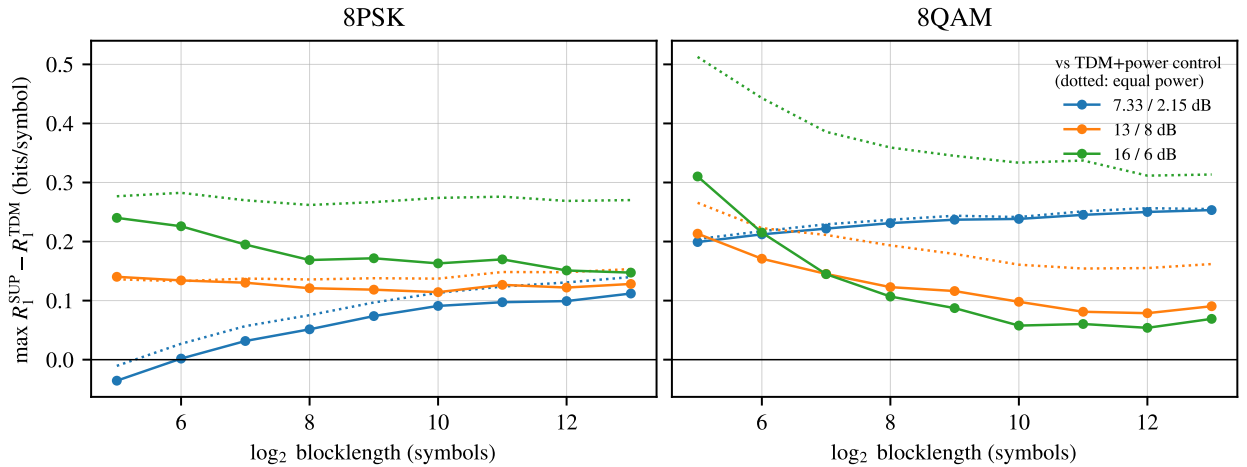


Figure 3: Largest R_1 advantage of the best 8-ary design over TDM with power control (solid) and equal-power TDM (dotted), against blocklength in symbols.

9 Bit-level polarization with unequal protection

Bit levels. By the chain rule, the symbol channel splits into binary level channels (multilevel coding [7, 8]):

$$I(X; Y) = I(b_2; Y) + I(b_1; Y | b_2) + I(b_0; Y | b_2, b_1).$$

Decoding each bit on its own (bit-interleaved coded modulation, BICM [9]) gives $\sum_b I(b; Y)$ instead. With Gray labels the two differ by at most 0.011 bits/symbol here: the levels are nearly independent but unequally protected (Table 8).

Polarization happens within a level. Arikan’s transform over n uses of one level channel sorts that level into good and bad synthetic channels, and the fraction of good ones tends to the level’s capacity [10]. It does not move protection between levels: at $n = 128$ the weak 8PSK level b_0 has 22 good synthetic channels and 73 still unpolarized, against 51 and 65 for b_2 and b_1 .

Two ways to code across levels. (i) One polar code over all levels: the labeling acts as additional polarization stages, and a length- $3n$ code chooses its frozen set from all $3n$ synthetic channels, each evaluated with its own level’s statistics [11, 12]. Polarization still holds for such non-identical channels [13], but channel-independent orderings such as the 5G NR sequence [14, 15] assume identical channels, so a level-aware construction (e.g., Gaussian approximation per level [16] or density evolution [17]) and a careful mapping of bits to levels are needed. (ii) One code per level with multistage decoding.

The broadcast bit split is option (ii) with levels assigned to users. For 8PSK the weak user’s code covers b_2b_1 and is decoded with b_0 as noise (1.567 bits/symbol at 8 dB); the strong user decodes b_2b_1 , then its own code on b_0 (0.934 at 13 dB). For 8QAM the weak user takes b_2 (0.811) and the strong user b_1b_0 given b_2 (1.910). All bits are uniform, so each code is an ordinary point-to-point polar code. On the BSBC there is only one level, and the only way to create a cloud and a satellite is to bias the satellite and carve both out of one polarization process, which is where the paper’s finite-length selection conflict comes from. With unequal levels the modulation supplies the split and that conflict disappears.

Table 8: Level capacities (multistage order), Bhattacharyya parameters Z , and how a length-128 polar transform polarizes each level (good/bad synthetic channels at tolerance 10^{-3} , from the erasure-channel recursion, an upper bound used for illustration). Hard decisions, 8 dB.

		b_2	$b_1 \mid b_2$	$b_0 \mid b_2b_1$
8PSK	capacity	0.784	0.784	0.583
	Z	0.30	0.30	0.55
	good / bad of 128	51 / 12	51 / 12	22 / 33
8QAM	capacity	0.811	0.625	0.621
	Z	0.26	0.51	0.52
	good / bad of 128	57 / 10	27 / 30	27 / 32

10 Fine rate control

A pure bit split gives only corner points. For 8PSK at 13/8 dB the modes are: the weak user takes all three bits ($R_2 \leq 2.150$), b_2b_1 ($R_2 \leq 1.567$, $R_1 \leq 0.934$), b_2 ($R_2 \leq 0.784$, $R_1 \leq 1.901$), or nothing ($R_1 \leq 2.868$). Lowering one user’s code rate moves inward only. Two knobs move along the boundary:

- **Mode mixing.** Use one split mode on a fraction μ of the symbols and the neighbouring mode on the rest, with one code per user spanning all of its bits. In 8PSK between “ $b_2b_1 \mid b_0$ ” and “ $b_2 \mid b_1b_0$ ”, the weak user’s code covers b_2 everywhere and b_1 on the first group; the strong user’s covers b_0 everywhere and b_1 on the second. Per-symbol means and dispersions are the μ -weighted averages of the two modes’, so the rate split moves continuously with μ (steps of $1/n$).
- **Biased level.** The paper’s construction on the one level that differs between neighbouring modes: the satellite flips that bit with probability $\alpha \in (0, \frac{1}{2})$.

Table 9 and Figure 4 show that mode mixing is the knob for short blocklengths: at $n = 128$ it is the best design found to within 0.012 bits/symbol and beats TDM with power control by up to about 0.13 bits/symbol in the middle of the region (8PSK at $R_2 = 1.32$: 0.855 against 0.722). The biased level has worse dispersion and loses up to 0.22 bits/symbol at $n = 128$; it matches the best design at some rate splits from $n \approx 2048$ and asymptotically, and it brings back the satellite-selection problem on that level. Practical points for mode mixing:

- Code lengths are generally not powers of two (in 8PSK the weak user’s code has $(1+\mu)n$ coded bits and the strong user’s $(2-\mu)n$), so rate matching by puncturing or shortening [14, 15] is needed, or values of μ that give power-of-two lengths.
- Each user’s code spans two kinds of bit channel, so its frozen set should come from per-channel reliabilities [12, 16].
- The choice of μ for a target (R_1, R_2) is a one-dimensional search with the mixed moments.

Table 9: Largest shortfall (bits/symbol) of each knob from the best design found, over the whole R_2 range, at 13/8 dB.

	$n = 128$	$n = 512$	$n = 2048$	capacity
mode mixing, 8PSK / 8QAM	0.006 / 0.012	0.022 / 0.023	0.020 / 0.047	0.062 / 0.109
biased level, 8PSK / 8QAM	0.212 / 0.222	0.115 / 0.115	0.073 / 0.073	0.079 / 0.102

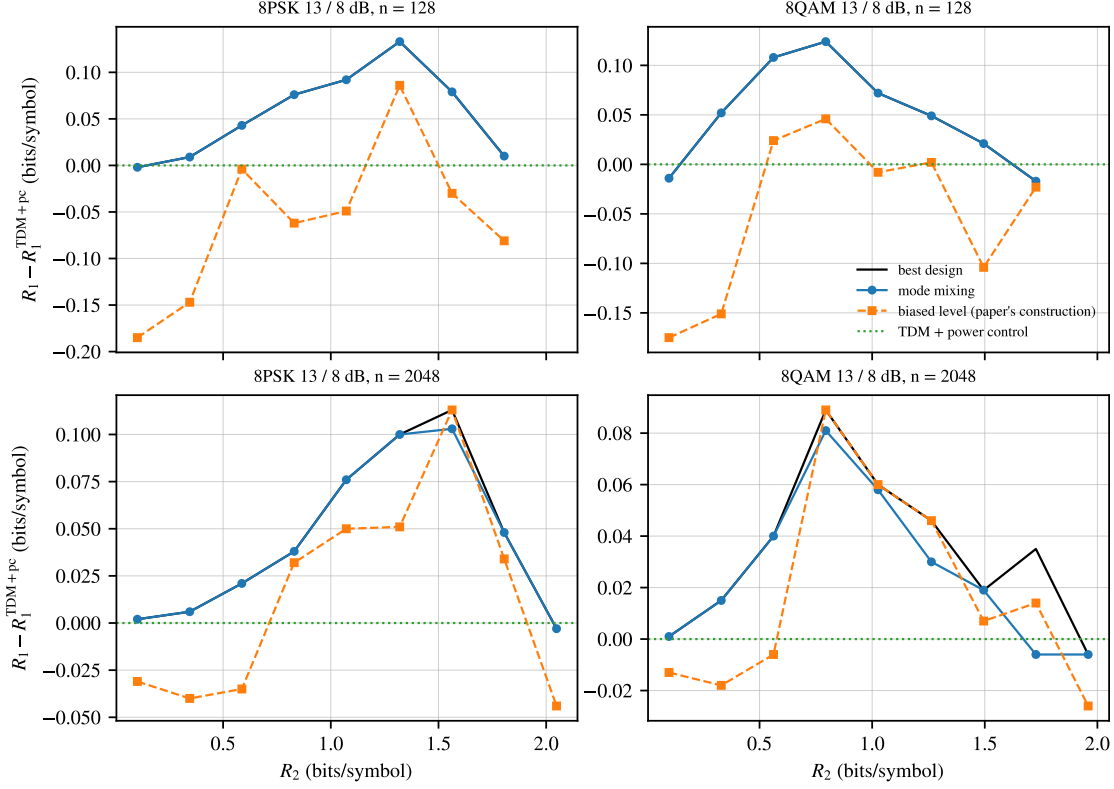


Figure 4: Advantage in R_1 over TDM with power control at 13/8 dB, for the best design, mode mixing, and the biased level, at $n = 128$ (top) and $n = 2048$ (bottom).

11 Polar-code simulations of the 8-ary schemes

Setup. Hard-decision 8PSK and rectangular 8QAM at 13/8 dB, $n = 128$ symbols, per-user block error rate $\leq 10^{-2}$, successive-cancellation (SC) decoding with $G_N = F^{\otimes m}$. For every code the information set comes from a genie-aided Monte-Carlo construction (20,000 blocks) on that code's own channel; the information sizes are the largest whose simulated block error rates, from 8,000 blocks, meet the target (binary search). Codes whose length is not a power of two are rate-matched by shortening (freeze and zero the last positions, which only depend on themselves under $F^{\otimes m}$) or by quasi-uniform puncturing [15]; each configuration was run both ways and the better is reported. Coded bits are spread over their (symbol, level) slots by a fixed random interleaver, and decoders use per-bit LLRs of the hard decisions with the other bits of the symbol either known (after SIC) or uniform.

- *Mode mixing*, $\mu \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ ($\mu = 1$: weak user $b_2 b_1$, strong user b_0 ; $\mu = 0$: weak user b_2 ,

Table 10: Polar mode mixing at $n = 128$, 13/8 dB: realized rates (bits/symbol), the margin over the upper convex hull of all simulated polar TDM points, and the loss to the mode-mixing normal approximation at the same R_2 .

μ	8PSK			8QAM		
	(R_2, R_1)	vs. TDM hull	loss to NA	(R_2, R_1)	vs. TDM hull	loss to NA
0	(0.523, 1.633)	+0.033	0.256	(0.562, 1.656)	+0.212	0.228
$\frac{1}{4}$	(0.641, 1.406)	+0.010	0.330	(0.633, 1.430)	+0.122	0.357
$\frac{1}{2}$	(0.805, 1.180)	+0.068	0.343	(0.727, 1.219)	+0.094	0.420
$\frac{3}{4}$	(0.961, 0.961)	+0.120	0.358	(0.820, 0.992)	+0.050	0.477
1	(1.133, 0.773)	+0.231	0.323	(0.945, 0.797)	+0.097	0.462

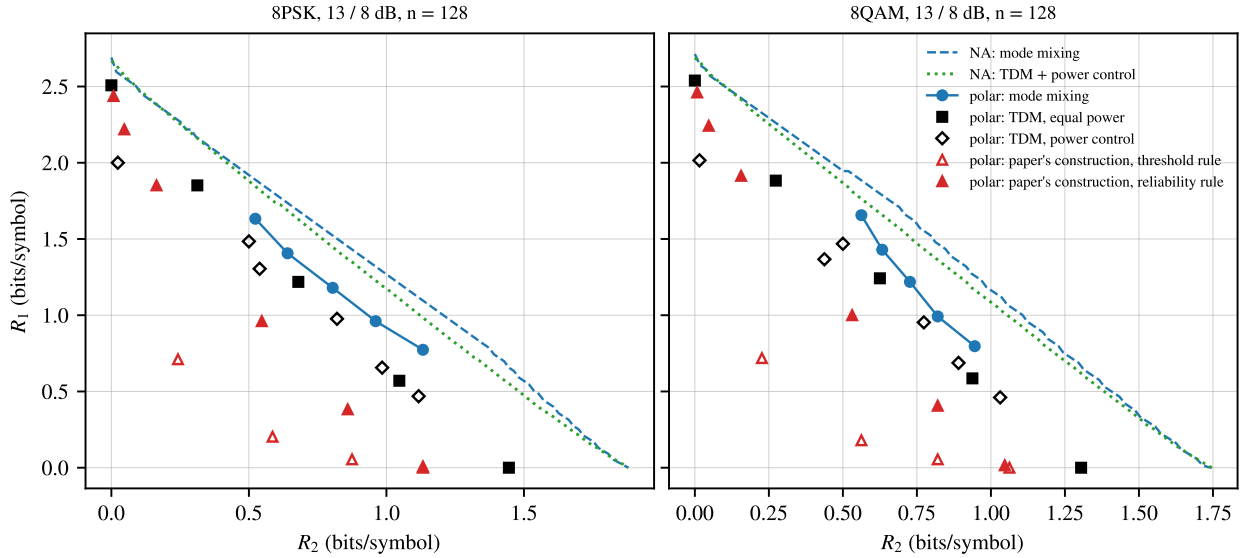


Figure 5: Polar codes at $n = 128$ (markers; SC decoding, per-user block error rate $\leq 10^{-2}$) against the normal approximations at the same blocklength (lines), hard-decision 8PSK and 8QAM at 13/8 dB.

strong user b_1b_0). One code per user; the strong user decodes the weak user's code, re-encodes it, and decodes its own bits given the weak user's bits in each symbol.

- *TDM*: user 1 alone, user 2 alone, and $\lambda \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ with $P_1 \in \{0.6, 1.0, 1.3\}$ (equal power at $P_1 = 1$); each user codes over all three bits of its symbols.
- *The paper's construction on every bit*, $a \in \{0.02, 0.05, 0.1, 0.2\}$: a cloud code over the $3n$ cloud bits and a satellite code whose frozen bits follow the deterministic rule ψ (an SC computation from the cloud at the encoder, repeated at the decoder). Satellite positions come from two genie-aided metrics: the error rate $e_H(i)$ of the most likely value of $U_1(i)$ given the cloud alone (near $\frac{1}{2}$ for a high-entropy position) and the error rate $e_L(i)$ given the cloud and the channel output. Two rules were tested: a common threshold on $\frac{1}{2} - e_H$ and e_L , the analogue of the paper's baseline (Algorithm 1); and a reliability rule that takes the positions with the smallest e_L among those with $e_H \geq \tau$, for $\tau \in \{0, 0.02, 0.05, 0.1, 0.2, 0.3, 0.4\}$.

Table 11: Paper’s construction on BSBC(0.01, 0.10), $n = 128$: the largest R_1 this simulator realizes with R_2 at least that of each thesis point (over $a \in \{0.02, \dots, 0.3\}$), against the thesis’s realized points read from its Fig. 4.

thesis point (R_2, R_1)	common-threshold rule	best reliability rule
(0.008, 0.375)	0.352	0.625
(0.055, 0.281)	0.039	0.367
(0.117, 0.180)	0.016	0.234
(0.180, 0.055)	0.000	0.000

Results. Table 10 and Figure 5. Polar mode mixing lies above the convex hull of every simulated TDM point, including the power-controlled ones, at all five rate splits, by 0.01–0.23 bits/symbol for 8PSK and 0.05–0.21 for 8QAM; the hull is generous to TDM, since it allows time sharing between configurations. The margin exceeds the normal-approximation prediction at $n = 128$ (+0.13 for 8PSK, +0.14 for 8QAM). Two effects contribute and were not separated: the TDM power grid has only three values, and the codes of the two schemes lose different amounts to the normal approximation (the TDM codes span all three unequally protected levels, the mode-mixing codes more homogeneous ones). The polar codes lose 0.23–0.48 bits/symbol to the normal approximation; the strong user alone loses 0.16 (8PSK) and 0.15 (8QAM).

The paper’s construction and the choice of satellite positions. With the common-threshold rule the satellite carries little: for 8PSK the construction realizes (1.13, 0) at $a = 0.02$ down to (0.24, 0.71) at $a = 0.2$, 0.47–1.38 bits/symbol below the TDM hull and far below its own normal approximation (about (0.90, 0.96) at $a = 0.1$). The rule weighs “random given the cloud” and “reliable given the channel” equally, so it rejects reliable positions that are only partly random in favour of fully random but unreliable ones, the failure the paper’s $n = 16$ example illustrates. The reliability rule fixes most of this: at $a = 0.1$ the 8PSK satellite rate rises from 0.20 to 0.96 bits/symbol, and at $a = 0.2$ from 0.71 to 1.85, with small values of τ (0–0.1) best. Even so, the construction stays below the TDM hull at every a tested: by 0.53, 0.61, 0.60, 0.37, 0.21 and 0.06 bits/symbol for 8PSK at $a = 0.02, 0.05, 0.1, 0.2, 0.3, 0.4$, and by 0.49, 0.54, 0.51, 0.32, 0.21 and 0.06 for 8QAM, the margin closing only as the cloud rate goes to zero. Mode mixing is above the hull at every rate split.

Validation on the paper’s BSBC. The same construction code, run on BSBC(0.01, 0.10) at $n = 128$ binary channel uses, can be compared with the thesis’s realized polar superposition points (Table 11). With the reliability rule the simulator matches or exceeds them except next to the weak user’s largest rate, where it realizes no satellite rate; with the common-threshold rule it falls well below them. The thesis’s points used the paper’s Algorithm 2 (5G NR ordering), which is not reimplemented here.

How good is the reliability rule? A fully exhaustive search over satellite sets is impossible at $n = 128$ (choosing about 100 of 512 positions has some 10^{110} possibilities), so the rule was checked in two ways. At $n = 16$ and 32 on the BSBC every satellite set of up to five (respectively three) positions was simulated, on the same random draws for all sets. At $n = 128$ a local search started from the best rule’s largest feasible set and added positions one at a time, or swapped out the position that most often started a decoding error, while both users still met 10^{-2} . Because

Table 12: Satellite sets of the paper’s construction: the best reliability rule against exhaustive search ($n = 16, 32$, BSBC) and local search ($n = 128$). “Positions” is the largest number of satellite information bits with both users at block error rate $\leq 10^{-2}$; the last columns are block error rates (strong, weak user) on fresh draws.

case	best rule (τ)	positions	search	positions	fresh-draw BLER: rule set / searched set
BSBC $n = 16, a = 0.1$	0.05	2	exhaustive	3	(0.0008, 0.0083) / (0.0024, 0.0084)
BSBC $n = 32, a = 0.1$	0.3	5	exhaustive (≤ 3 searched)	≥ 3	(0.0097, 0.0081) / –
BSBC $n = 32, a = 0.2$	0.3	5	exhaustive (≤ 3 searched)	≥ 3	(0.0021, 0.0021) / –
BSBC $n = 128, a = 0.1$	0.4	15	local	23	(0.0100, 0.0040) / (0.0095, 0.0089)
BSBC $n = 128, a = 0.2$	0.2	60	local	64	(0.0096, 0.0028) / (0.0107, 0.0109)
8PSK $n = 128, a = 0.1$	0.2	106	local	112	(0.0114, 0.0038) / (0.0114, 0.0111)
8PSK $n = 128, a = 0.2$	0.2	199	local	203	(0.0091, 0.0024) / (0.0100, 0.0086)

choosing the best of many sets on fixed draws flatters the winner, every winning set and every rule’s set was re-checked on 20,000 fresh, independent blocks (Table 12).

The best reliability rule is close to the best sets found. At $n = 16$ it is one position short of the exhaustive optimum; at $n = 32$ it reaches more positions than the exhaustive search could cover; and in the exhaustive runs its sets rank between 1st and 45th among up to 4,960 sets, against 10th to 1,079th for the common-threshold rule. At $n = 128$ the local search adds 4–8 positions (0.03–0.06 bits per channel use), but these sets sit at the target on fresh draws and two of the four slightly exceed it (by one to two standard errors): they use up the weak user’s error margin, which the rule’s sets leave unused, so the true gain is smaller. The best τ varies (0.05 at $n = 16$, 0.2–0.4 at $n = 32$ and 128), so a one-dimensional sweep over τ is advisable for each case. Better satellite positions therefore recover at most about 0.05 bits/symbol, against a 0.37–0.6 bits/symbol gap to TDM in the middle of the 8PSK region: that gap is a property of the biased-satellite construction at this length, not of the selection rule. (The $n = 16, a = 0.2$ case was also searched but supports no cloud bits, so it is not a superposition case.)

12 Discussion and implications for the paper

- **The conclusion’s conjecture needs qualifying.** Larger alphabets do not help in the paper’s sense when the symbol has 2 bits: the discrete construction needs the same or longer blocklength, and the power-domain gains are power allocation that orthogonal access also obtains. With 3-bit symbols and unequal bit protection they do help, but through bit splitting, not through the paper’s biased satellite.
- **A possible reframing.** “When does superposition pay off at finite blocklength?” has a clean answer across the models here: when the alphabet offers unequally protected components that can be assigned to the users. Otherwise orthogonal access is as good or better at short blocklengths, consistent with [4].
- **Polar implementation.** In the regime where superposition pays off, the layers carry uniform bits and each is an ordinary point-to-point polar code (multilevel polar coding with SIC [11]). The paper’s construction remains useful as a long-blocklength refinement on one level.
- **The polar simulations support the conclusion.** At $n = 128$ polar mode mixing beats polar TDM at every rate split tested. The paper’s biased-satellite construction stays below both, by up to 0.6 bits/symbol even with a good selection rule, and its performance hinges on the choice of satellite positions.

- **Next experiments.** Soft-decision 8PSK/8QAM against TDM with power control (the 4-point soft results suggest the margin may shrink); CRC-aided list decoding [18] for all schemes; a finer TDM power grid; other SNR pairs and blocklengths; star 8QAM; and 16-point hierarchical constellations.

13 Limitations

- Normal approximations without third-order terms; values at $n \leq 64$ are indicative only. For the BSBC, adding $\frac{1}{2} \log n/n$ to every constituent code did not move the TDM crossover.
- The 8PSK and 8QAM results are hard-decision models. Their polar simulations cover one SNR pair (13/8 dB) and $n = 128$, with SC decoding only, a three-value TDM power grid, block error rates from 8,000 blocks (about $\pm 11\%$ relative at 10^{-2} , one standard deviation), and, for the paper’s construction, the selection rules described in Section 11 rather than its Algorithm 2. Exhaustive searches of satellite sets cover only $n \leq 32$ and up to five positions; at $n = 128$ only a local search was possible.
- 4-point constellations are the parallelogram family $a(v) + b(s)$; the 8QAM is the rectangular grid only; the quaternary discrete channel is either the QSC or hard-decision Gray QPSK.
- TDM with power control assumes an average power constraint across a user’s slots and the same constellation and detector in every slot.
- The design grids are coarse (64 values of β ; 8 values per bit for 8-ary). Random and local searches show the 8-ary gains are conservative by up to about 0.1 bits/symbol asymptotically.
- The 4-point AWGN polar simulations: one SNR pair, $n = 128$, SC decoding, block error rates from 6,000 blocks (about $\pm 25\%$ relative at 10^{-2}), rate resolution $1/128$.

14 Reproducibility

All scripts are in `scripts/` and need only numpy and matplotlib; the longer sweeps were run on an AWS c7i instance.

- `qam_bc.py`, `qam_sweep.py`: 4-point AWGN normal approximations (`qam_sweep_results.txt`).
- `qam_polar_sim.py`, `qam_realized.py`: polar encoder, SC decoder, genie construction and rate search (`qam_realized_7.33dB_2.15dB_n128.txt`).
- `qsbc.py`, `qsbc_sweep.py`: QSBC moments, families, random search and normal approximations (`qsbc_sweep_results.txt`); `qsbc.py` also provides the general-matrix routines used below.
- `gray_qpsk_bc.py`: hard-decision Gray QPSK (`gray_qpsk_results.txt`).
- `hard_8ary_bc.py`, `hard_8ary_pc.py`: 8PSK/8QAM channel matrices, designs, comparisons with equal-power and power-controlled TDM (`hard8_*.txt`, `hard8pc_*.txt`).
- `bit_levels_8ary.py`: Table 8 (`bit_levels_results.txt`). `rate_control_8ary.py`: Section 10 (`ratectl*_13_8.txt`).

- `polar_8ary_sim.py` (with `run_polar_8ary.sh` for all 72 configurations, about 10 minutes on 14 cores): Section 11 (`polar_8ary_results.jsonl`); `na_curves_8ary.py`: the reference curves (`na_curves_8ary.csv`); `polar_8ary_summary.py`: Tables 10 and 11 (`polar_8ary_summary.txt`). The selection-rule runs use `polar_8ary_sim.py --select` (`gag_select_results.jsonl`) and `gag_bsbc_sim.py` (`gag_bsbc_results.jsonl`); Table 12 comes from `gag_search.py` (`gag_search_results.jsonl`).
- `sup_vs_ccp.py`: Table 5. `interior_gaps.py`: Figure 2 and Table 2.
- `make_report_figures.py`: all figures.

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